

SightMate: A Vision-Based Assistive Navigation System for the Visually Impaired

Deepak R V
UG Scholar

Department of CSE(AIML)
S. A. Engineering College
Chennai, India
deepakvetrivelan@gmail.com

Gowtham S
UG Scholar

Department of CSE(AIML)
S. A. Engineering College
Chennai, India
2227018@saec.ac.in

Harmithaa V
UG Scholar

Department of CSE(AIML)
S. A. Engineering College
Chennai, India
2227019@saec.ac.in

Mrs. V. Gnanapriya
Assistant Professor

Department of CSE(AIML)
S. A. Engineering College
Chennai, India
gnanapriyap@saec.ac.in

Abstract- Visual impairment significantly restricts independent mobility and environmental awareness for millions of individuals worldwide, creating challenges in safely navigating unfamiliar environments. Traditional assistive tools such as white canes and guide dogs provide limited perception of distant obstacles and complex surroundings. Recent advances in computer vision and mobile artificial intelligence have enabled the development of intelligent assistive technologies capable of interpreting environmental scenes in real time. This paper presents SightMate, an AI-driven vision-based assistive navigation system designed to support visually impaired users through real-time environmental perception and auditory guidance. The proposed system integrates multiple perception and interaction layers including camera-based scene acquisition, obstacle detection using the YOLOv8 deep learning object detection framework [4], and walkable path segmentation using the Fast-SCNN semantic segmentation network [5] to identify safe navigation regions. A zone-based navigation decision module analyzes spatial obstacle distribution across left, center, and right regions of the captured frame to generate intuitive movement instructions. To improve robustness in dynamic environments, motion filtering and frame stability analysis are incorporated to reduce false detections caused by camera movement. The system also includes accessibility features such as voice-command interaction, Optical Character Recognition (OCR) for environmental text reading using ML Kit [15], and a six-dot Braille keyboard interface that enables tactile user input. The complete solution is implemented as a Flutter-based mobile application [21] utilizing TensorFlow Lite [14] for on-device inference, enabling low-latency real-time processing without dependence on continuous internet connectivity. Experimental evaluation demonstrates reliable obstacle detection and walkable path identification under diverse environmental conditions while maintaining computational efficiency suitable for smartphone deployment. The proposed framework bridges the gap between mobile computer vision research and practical assistive technology by delivering a scalable, real-time, and user-centric navigation assistance platform for visually impaired individuals.

Keywords— Assistive Navigation, Visual Impairment, YOLOv8, Fast-SCNN, Object Detection, Semantic Segmentation, OCR, Braille Interface, Voice Assistance, TensorFlow Lite, Mobile AI, Computer Vision.

Visual impairment represents a major global public health challenge, affecting millions of individuals who rely on assistive technologies to navigate their surroundings safely and independently. According to the World Health Organization (WHO), more than 2.2 billion people worldwide experience some form of vision impairment, with a significant proportion facing severe difficulties in performing everyday navigation tasks [1]. For visually impaired individuals, independent mobility in unfamiliar environments remains one of the most difficult aspects of daily life. Traditional assistive aids such as white canes and guide dogs provide essential tactile feedback but are limited in their ability to detect distant obstacles, understand environmental context, or interpret complex surroundings such as crowded public spaces, signage, and dynamic obstacles.

With the rapid advancement of mobile computing and artificial intelligence, smartphones have emerged as powerful platforms capable of performing sophisticated visual perception tasks in real time. Modern smartphones are equipped with high-resolution cameras, powerful processors, and machine learning acceleration capabilities, enabling the deployment of deep learning models directly on mobile devices. These technological advancements have created new opportunities for developing intelligent assistive systems that can analyze visual environments and provide contextual feedback to visually impaired users.

Recent research in computer vision has demonstrated the effectiveness of deep learning models for scene understanding tasks such as object detection and semantic segmentation. Object detection algorithms, particularly the You Only Look Once (YOLO) family of models, have demonstrated high accuracy while maintaining real-time inference performance [2], [4], [10]. Similarly, semantic segmentation models enable pixel-level classification of images, allowing systems to distinguish between walkable surfaces, obstacles, and environmental structures [5], [24]. When combined, these techniques provide a more comprehensive understanding of the surrounding environment, which is essential for reliable navigation assistance.

I. INTRODUCTION

Several studies have explored vision-based navigation systems designed specifically for visually impaired users. These systems typically rely on camera input combined with machine learning techniques to identify obstacles and guide users through complex environments [16], [17], [22]. However, many existing approaches rely on specialized hardware, limited environmental perception, or computationally expensive models that are difficult to deploy on resource-constrained mobile devices. Consequently, there remains a need for lightweight assistive systems capable of performing reliable real-time scene analysis using widely available smartphones.

To address these challenges, this paper presents **SightMate**, a lightweight vision-based assistive navigation system designed to support visually impaired individuals through real-time environmental perception and audio guidance. The proposed system integrates multiple perception and interaction modules to transform camera input into actionable navigation instructions. Specifically, the system employs **YOLOv8-based object detection** to identify potential obstacles in the user's path [4] and **Fast-SCNN semantic segmentation** to detect walkable regions within the scene [5]. A zone-based navigation decision algorithm processes the outputs of these models by dividing the camera frame into left, center, and right regions, enabling the system to generate intuitive movement instructions such as "move left," "move right," or "path clear."

To enhance reliability in dynamic environments, the system incorporates motion filtering techniques that analyze frame-to-frame variations to reduce false detections caused by camera movement or temporary environmental changes. In addition to navigation assistance, the proposed system integrates several accessibility features designed to improve usability for visually impaired users. These include **voice-based command interaction**, **Optical Character Recognition (OCR)** for reading printed text from the environment using ML Kit [15], and a **six-dot Braille keyboard interface** implemented using multi-touch input, enabling tactile interaction with the application.

The entire system is implemented as a **Flutter-based mobile application** [21] with **TensorFlow Lite** enabling efficient on-device inference of deep learning models [14]. This architecture allows the application to operate with minimal latency while avoiding dependence on continuous internet connectivity, making it suitable for real-world deployment on widely available smartphones.

II. LITERATURE SURVEY

Recent advances in computer vision and mobile artificial intelligence have enabled the development of assistive technologies aimed at improving independent mobility for visually impaired individuals. Early navigation assistance systems primarily relied on sensor-based approaches such as ultrasonic sensors and wearable devices to detect nearby obstacles. Although these systems provided basic obstacle awareness, they lacked the ability to interpret complex

environmental scenes or identify safe navigation paths in dynamic environments.

With the emergence of deep learning, vision-based assistive navigation systems have gained significant attention. Object detection models such as the **You Only Look Once (YOLO)** family have demonstrated high performance for real-time object detection tasks in complex environments [2], [10]. These models are capable of detecting multiple objects simultaneously while maintaining low inference latency, making them suitable for real-time applications deployed on mobile devices. Recent implementations such as **YOLOv5** and **YOLOv8** have further improved detection accuracy and efficiency through architectural optimizations and improved training strategies [3], [4].

In addition to object detection, **semantic segmentation** has become an important technique for understanding environmental context. Semantic segmentation models classify each pixel of an image, enabling systems to differentiate between walkable areas, obstacles, and environmental structures. Models such as **Fast-SCNN** have been specifically designed for efficient real-time segmentation on mobile devices, making them particularly suitable for assistive navigation applications [5]. Other segmentation architectures such as **U-Net** and **Pyramid Scene Parsing Network (PSPNet)** have also demonstrated strong performance in scene understanding tasks [7], [25]. Recent research has further explored lightweight segmentation techniques optimized for edge devices and mobile platforms [24].

Several research studies have proposed vision-based navigation systems specifically for visually impaired users. Lakde et al. [16] proposed a deep learning-based obstacle detection system capable of identifying objects in real-time environments using camera input. Similarly, Al-Khalifa and Al-Razgan [17] developed a smartphone-based assistive navigation system that utilizes camera data to guide visually impaired users through voice feedback. Patel and Shah [22] introduced a deep learning-based mobile navigation assistant capable of detecting obstacles and providing directional guidance through smartphone applications. These systems demonstrate the feasibility of using mobile computer vision technologies to improve accessibility and independent mobility.

Recent developments in **Edge AI and mobile computing** have further facilitated the deployment of deep learning models on smartphones. Lightweight neural network architectures such as **MobileNet** have been designed specifically for resource-constrained devices, enabling efficient inference while maintaining competitive accuracy [8], [9]. Frameworks such as **TensorFlow Lite** enable optimized deployment of machine learning models on mobile platforms, supporting real-time inference directly on smartphones without requiring cloud connectivity [14]. Additionally, mobile development frameworks such as **Flutter** allow developers to build cross-platform applications that integrate computer vision models with user interaction modules and accessibility features [21].

Despite these advancements, many existing systems still face limitations related to computational efficiency,

environmental robustness, and user interaction. Some approaches rely on specialized hardware or require continuous internet connectivity, which limits their practicality in real-world scenarios. Furthermore, many systems focus solely on obstacle detection without incorporating additional accessibility features such as text recognition, voice interaction, or alternative input methods.

To address these limitations, the proposed **SightMate system** integrates multiple complementary technologies including **YOLOv8-based object detection**, **Fast-SCNN semantic segmentation**, and a **zone-based navigation decision algorithm** to generate real-time movement instructions for visually impaired users. In addition to navigation assistance, the system incorporates accessibility features such as **Optical Character Recognition (OCR)**, **voice-command interaction**, and a **six-dot Braille keyboard interface**, creating a comprehensive mobile assistive platform designed for real-world deployment.

III. SYSTEM ARCHITECTURE

The proposed **SightMate system** follows a modular architecture designed to provide real-time navigation assistance for visually impaired users using a smartphone camera and on-device artificial intelligence models. The architecture integrates multiple perception, decision-making, and interaction components to transform visual input into meaningful auditory guidance. Fig. 1 illustrates the overall system architecture and the interaction between the different processing modules.

The system begins with **camera-based scene acquisition**, where the smartphone camera continuously captures frames from the surrounding environment. These frames serve as the primary input to the computer vision pipeline. Each captured frame is preprocessed and forwarded to the object detection and scene understanding modules for further analysis.

The first major processing stage is the **object detection module**, which employs the YOLOv8 deep learning framework to identify potential obstacles present in the environment [4]. YOLO-based models are widely used for real-time detection due to their ability to perform object localization and classification in a single neural network pass [2], [10]. The model detects objects such as pedestrians, vehicles, furniture, and other environmental hazards that may obstruct the user's path.

In parallel with object detection, the system performs **semantic segmentation** using the Fast-SCNN network. Semantic segmentation assigns a class label to every pixel in the image, allowing the system to distinguish between walkable areas and non-walkable regions [5]. Fast-SCNN is specifically designed for mobile and embedded platforms, enabling efficient inference while maintaining acceptable segmentation accuracy. This capability allows the system to identify safe walking paths within the environment.

The outputs from the object detection and segmentation modules are then processed by the **navigation decision module**. In this stage, the camera frame is divided into three spatial regions: **left**, **center**, and **right**. The system analyzes the spatial distribution of detected obstacles and segmented walkable areas across these regions. Based on this analysis,

the navigation algorithm generates movement instructions such as *"move left," "move right," or "path clear."* This zone-based strategy simplifies navigation instructions and ensures that the user receives intuitive and actionable guidance.

To improve system reliability in dynamic environments, a **motion filtering component** is incorporated into the architecture. Motion filtering compares consecutive frames to detect sudden environmental changes and reduce false detections caused by camera movement or temporary occlusions. This mechanism stabilizes the perception pipeline and ensures consistent navigation feedback.

In addition to navigation assistance, the system integrates several **accessibility and interaction modules** to enhance usability. A **voice command module** allows users to control application functions through speech interaction. An **Optical Character Recognition (OCR) module** based on Google ML Kit enables the system to detect and read printed text from the environment, converting recognized text into speech output [15]. Furthermore, a **six-dot Braille keyboard interface** implemented using multi-touch input allows visually impaired users to enter text commands using tactile interaction.

All processing components are integrated within a **Flutter-based mobile application architecture** [21]. The deep learning models used for object detection and segmentation are deployed using **TensorFlow Lite**, enabling optimized on-device inference and reducing computational overhead [14]. By performing inference locally on the smartphone, the system eliminates the need for continuous internet connectivity and significantly reduces response latency.

Through the integration of these modules, the SightMate architecture provides a lightweight, scalable, and efficient assistive navigation platform capable of operating in real-world environments using widely available smartphone hardware.

To provide useful contextual information during navigation, the system architecture also integrates a **device information module** that retrieves real-time system data such as battery level, current time, and date. This information is communicated to the user through voice output when requested, ensuring that important device and environmental information remains accessible during navigation.

These interaction modules operate alongside the core computer vision pipeline, forming a comprehensive assistive framework that combines perception, navigation guidance, and accessible user interaction within a unified mobile application architecture.

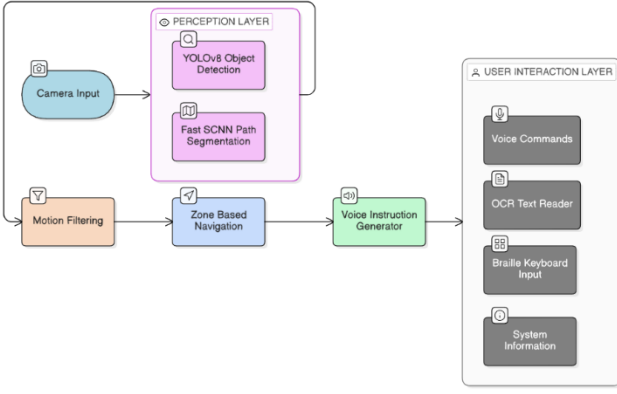


Fig. 1. System architecture of the proposed SightMate assistive navigation system.

A. Vision-Based Navigation Pipeline

Figure 2 illustrates the navigation decision workflow used to convert environment perception into real-time movement guidance.

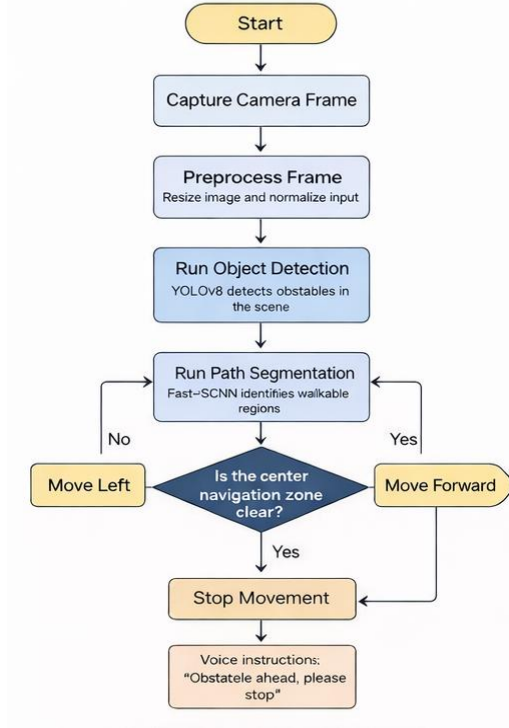


Fig. 2: Navigation decision flowchart.

B. Object Detection Pipeline

The object detection module is responsible for identifying obstacles present in the user's surrounding environment. As illustrated in Fig. 3, the captured camera frame is first preprocessed through resizing and normalization before being passed to the YOLOv8 deep learning model. The backbone network extracts hierarchical feature representations, while the neck module performs multi-scale feature fusion to improve detection accuracy for

objects of different sizes. The detection head then predicts bounding boxes and class labels for obstacles such as pedestrians, vehicles, bicycles, and street objects. Finally, post-processing using Non-Maximum Suppression removes redundant detections and produces the final obstacle predictions used by the navigation decision module.

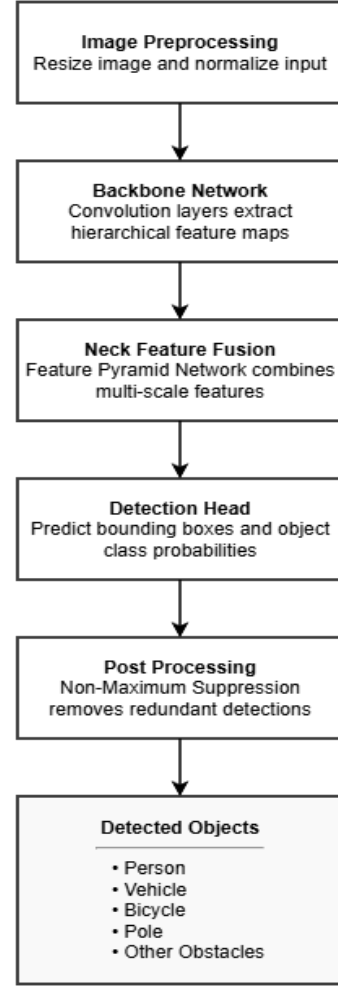


Fig. 3: YOLOv8 object detection pipeline.

C. Path Segmentation Workflow

The path segmentation module is responsible for identifying walkable regions within the surrounding environment. As illustrated in Fig. 4, the captured camera frame is processed using the Fast-SCNN semantic segmentation network, which is specifically designed for real-time performance on mobile devices. The network begins with a learning-to-downsample module that efficiently reduces the spatial resolution of the input image while preserving important visual features. The global feature extractor then captures contextual information using a lightweight convolutional encoder.

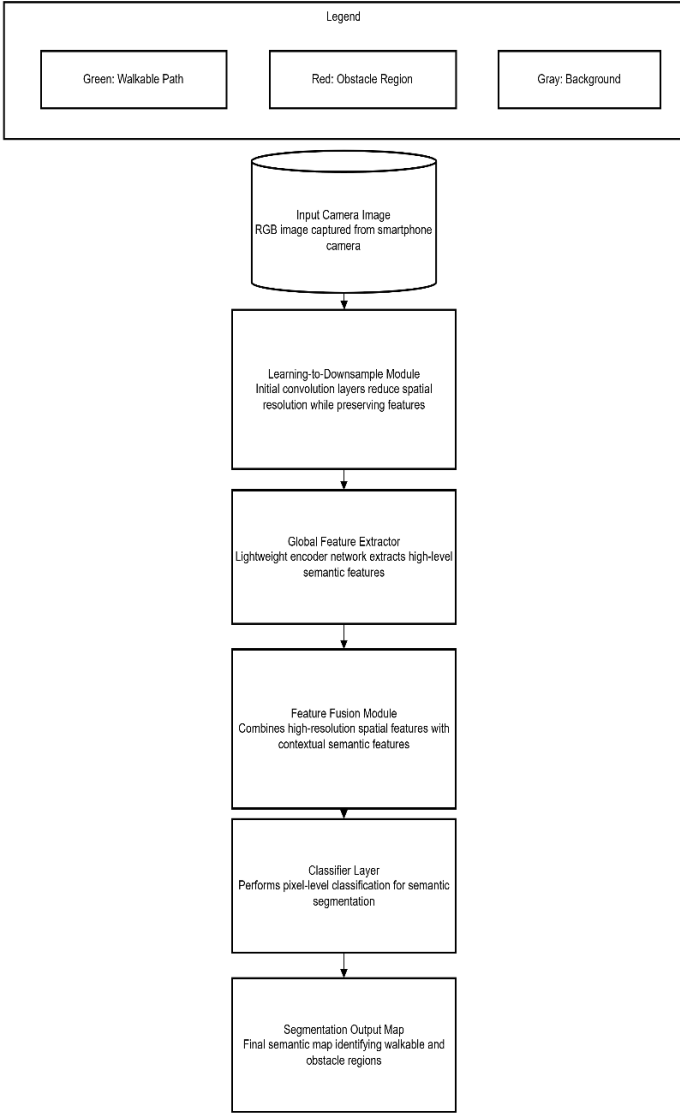


Fig. 4. Fast-SCNN semantic segmentation workflow.

IV. IMPLEMENTATION

A. Object Detection Model (YOLO)

The obstacle detection module uses the YOLO-based object detection framework, which predicts bounding boxes and class probabilities directly from the input image. For each detected object, the model predicts a confidence score representing the probability that an object exists within a predicted bounding box.

$$P(\text{Object}) = P(\text{Class} \mid \text{Object}) \times P(\text{Objectness}) \quad (1)$$

Where:

$P(\text{Object})$ represents the final detection confidence

$P(\text{Class} \mid \text{Object})$ denotes the conditional probability of the predicted object class

$P(\text{Objectness})$ indicates the likelihood that an object is present in the bounding box

Objects whose confidence score exceeds a predefined threshold are considered valid detections and forwarded to the navigation decision module.

B. Semantic Segmentation

The Fast-SCNN model performs pixel-level classification to determine whether each pixel belongs to a walkable path or an obstacle region. The segmentation output is represented as a probability map:

$$S(x,y) = \arg \max_{c \in C} P(c \mid x,y) \quad (2)$$

Where:

$S(x,y)$ represents the predicted class label for pixel location (x,y)

C is the set of semantic classes

$P(c \mid x,y)$ is the predicted probability that pixel (x,y) belongs to class c

In the proposed system, the segmentation output distinguishes between walkable paths, obstacles, and background regions.

C. Zone-Based Navigation Decision

The navigation decision module divides the camera frame into three spatial regions: left, center, and right. The safe navigation direction is determined by analyzing obstacle density within each region.

$$D_i = \sum_{k=1}^n A_k \quad (3)$$

Where:

D_i represents the obstacle density within region i

A_k represents the area of detected obstacle k

n is the number of detected obstacles in that region

The navigation direction is selected as:

$$\text{Direction} = \arg \min (D_{\text{left}}, D_{\text{center}}, D_{\text{right}}) \quad (4)$$

The system then generates corresponding voice instructions such as “move left”, “move right”, or “path clear.”

D. Motion Filtering

To reduce false detections caused by camera movement, frame-to-frame motion stability is analyzed using temporal filtering.

$$M_t = |F_t - F_{t-1}| \quad (5)$$

Where:

F_t represents the current frame

F_{t-1} represents the previous frame

V. EXPERIMENTAL SETUP AND EVALUATION

A. Hardware and Software Environment

The proposed SightMate system was implemented and evaluated on a smartphone-based mobile platform. The application was developed using the Flutter framework, enabling efficient cross-platform mobile deployment while providing direct access to device hardware components such as the camera, microphone, and touchscreen interface.

The deep learning models used in the system were deployed using TensorFlow Lite, which enables optimized on-device inference for mobile and embedded environments. The YOLOv8 object detection model and Fast-SCNN semantic segmentation model were converted into TensorFlow Lite format to reduce computational overhead and ensure real-time processing capability on mobile hardware.

All experiments were conducted using an Android smartphone equipped with a mid-range mobile processor, integrated GPU acceleration, and a high-resolution camera. The device continuously captured video frames from the environment, which were processed by the computer vision pipeline in real time.

B. Test Environment and Dataset

The system was evaluated using real-world environmental scenes captured through the smartphone camera under different indoor and outdoor conditions. The evaluation environments included pedestrian walkways, campus corridors, building entrances, and urban street pathways. The test dataset consisted of multiple video sequences recorded in various lighting conditions including daylight, indoor lighting, and moderate shadow conditions. These scenes contained common obstacles such as pedestrians, vehicles, bicycles, walls, furniture, and other environmental structures that may obstruct navigation. The objective of the evaluation was to verify the ability of the system to detect obstacles accurately, identify safe walking paths, and generate reliable navigation instructions for visually impaired users.

C. Evaluation Metrics

To assess the performance of the proposed system, several evaluation metrics commonly used in computer vision applications were considered.

For the object detection module, performance was evaluated based on detection accuracy and Intersection over Union (IoU) between predicted bounding boxes and ground truth object annotations. The IoU metric measures the overlap between the predicted bounding box and the actual object region, providing an indication of localization accuracy.

For the semantic segmentation module, pixel-level classification accuracy was used to measure the ability of the Fast-SCNN model to correctly distinguish walkable paths from obstacle regions and background areas. In addition to model accuracy, system performance was evaluated in terms of real-time responsiveness. Frame processing time and system latency were measured to ensure that navigation instructions could be generated promptly without noticeable delay. These metrics are essential for determining the practical usability of the system in real-world navigation scenarios.

D. Experimental Procedure

During evaluation, the smartphone camera continuously captured frames from the surrounding environment while the application processed the frames using the integrated computer vision pipeline. The YOLOv8 model detected objects within the scene, while the Fast-SCNN network generated segmentation maps identifying walkable paths. The navigation decision module then analyzed the spatial distribution of obstacles and safe regions within the frame. Based on the calculated obstacle density across the left, center, and right navigation zones, the system generated directional instructions such as “move left,” “move right,” or “path clear.”

These instructions were delivered to the user through the text-to-speech engine, providing real-time auditory guidance during navigation.

VI. RESULTS AND PERFORMANCE ANALYSIS

A. Obstacle Detection Performance

The performance of the YOLOv8 object detection module was evaluated across multiple environmental scenarios containing common obstacles such as pedestrians, vehicles, bicycles, and stationary objects. The model demonstrated strong detection capability across both indoor and outdoor environments.

The detection accuracy was evaluated using Intersection over Union (IoU) and detection confidence thresholds. The YOLOv8 model successfully detected the majority of obstacles within the camera field of view while maintaining real-time inference speed suitable for mobile devices. The model showed particularly strong performance in detecting pedestrians and large obstacles that commonly appear in pedestrian navigation environments.

B. Walkable Path Segmentation Results

The Fast-SCNN semantic segmentation model was evaluated based on its ability to correctly identify walkable paths within the captured scene. The segmentation network successfully differentiated between walkable surfaces and non-walkable regions such as walls, vehicles, and environmental obstacles.

The segmentation output produced pixel-level classification maps that highlighted safe navigation areas within the scene. These segmentation maps were then used by the navigation decision module to determine safe movement directions for the user.

The lightweight design of Fast-SCNN enabled efficient inference on mobile hardware while maintaining acceptable segmentation accuracy.

C. Navigation Guidance Performance

The zone-based navigation decision algorithm demonstrated reliable performance in generating movement instructions based on the spatial distribution of obstacles and walkable regions. By dividing the camera frame into left, center, and right navigation zones, the system was able to determine safe movement directions in real time.

When obstacles were detected in the center navigation zone, the system successfully identified alternative directions and generated appropriate instructions such as “move left” or “move right.” If no significant obstacles were detected, the system provided the instruction “path clear.”

This simplified navigation strategy enabled the system to deliver intuitive and easily understandable guidance for visually impaired users.

D. System Performance on Mobile Devices

The performance of the SightMate system was also evaluated in terms of processing latency and real-time responsiveness on smartphone hardware. The integration of TensorFlow Lite enabled efficient execution of both the YOLOv8 object detection and Fast-SCNN segmentation models on mobile devices.

The system was able to process camera frames in near real time, ensuring that navigation instructions were delivered with minimal delay. The lightweight architecture of the perception pipeline allowed the application to operate smoothly without excessive computational load on the device.

Overall, the results demonstrate that the proposed SightMate system can effectively perform real-time environmental perception and navigation assistance using widely available smartphone hardware.

VII. DISCUSSION

The experimental results demonstrate that the proposed SightMate system is capable of performing reliable real-time environmental perception using only smartphone hardware. The integration of object detection and semantic segmentation enables the system to analyze complex environments more effectively than approaches that rely on a single perception method.

The YOLOv8 object detection module provides accurate identification of dynamic and static obstacles, while the Fast-SCNN segmentation network identifies continuous walkable regions within the scene. The combination of these two perception mechanisms allows the system to generate more reliable navigation guidance for visually impaired users.

Another important contribution of the proposed system is the zone-based navigation strategy. Instead of providing complex spatial descriptions of the environment, the system simplifies navigation instructions into intuitive commands such as “move left,” “move right,” or “path clear.” This design significantly reduces cognitive load for users and enables faster decision-making during navigation.

The integration of accessibility features such as OCR-based text reading, voice command interaction, and a Braille keyboard interface further enhances the usability of the system. These modules allow visually impaired users to interact with the application through multiple input modalities, making the system more inclusive and practical for real-world deployment.

VIII. LIMITATIONS

Although the SightMate system demonstrates promising performance, several limitations remain. First, the accuracy of the object detection and segmentation models may decrease in extremely low-light conditions or environments with heavy visual clutter. In such scenarios, the camera input may not provide sufficient visual information for reliable scene interpretation.

Another limitation is related to camera motion. Rapid movement of the smartphone camera may introduce motion blur or unstable frames, which can affect detection accuracy. While the proposed motion filtering module reduces some of these effects, further improvements in stabilization techniques may enhance system robustness.

Additionally, the current system focuses primarily on obstacle detection and walkable path identification. More complex environmental understanding tasks, such as recognizing traffic signals, stairs, or specific navigation landmarks, are not yet incorporated into the current implementation.

IX. FUTURE WORK

Future research will focus on improving the environmental perception capabilities of the SightMate system. One possible enhancement involves integrating depth estimation models that can estimate the distance between the user and surrounding obstacles, enabling more precise navigation guidance.

Another potential direction involves incorporating reinforcement learning-based navigation strategies capable of adapting to dynamic environments and user behavior. Such approaches could allow the system to learn optimal navigation decisions based on continuous user feedback.

Further improvements may also involve expanding the OCR module to support multilingual text recognition and improving speech interaction capabilities through advanced natural language processing models. Additionally, integrating wearable devices such as smart glasses could provide a more ergonomic and hands-free assistive experience.

Future improvements may include integrating advanced scene understanding techniques to recognize environmental cues such as traffic signals, pedestrian crossings, stairs, and entrances. Incorporating this contextual information would enable the system to provide more intelligent and situation-aware navigation guidance. In addition, future work may focus on optimizing edge AI performance to reduce computational latency and energy consumption on mobile devices, improving battery efficiency and enabling longer real-time operation in practical environments.

X. CONCLUSION

This paper presented SightMate, a lightweight vision-based assistive navigation system designed to support visually impaired individuals through real-time environmental perception and audio guidance. The system integrates YOLOv8 object detection, Fast-SCNN semantic segmentation, and a zone-based navigation decision algorithm to analyze camera input and generate intuitive movement instructions.

The proposed system also incorporates several accessibility features including OCR-based text recognition, voice command interaction, and a Braille keyboard interface to improve usability for visually impaired users. By leveraging TensorFlow Lite for on-device inference and Flutter for mobile application development, the system achieves efficient real-time performance without requiring continuous internet connectivity.

Experimental evaluation demonstrates that the SightMate system can effectively detect obstacles, identify walkable paths, and provide reliable navigation guidance using widely available smartphone hardware. The proposed framework represents an important step toward developing intelligent assistive technologies that improve mobility, independence, and accessibility for visually impaired individuals.

REFERENCES

- [1] World Health Organization, "World report on vision," Geneva, Switzerland: WHO, 2019.
- [2] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," arXiv:1804.02767, 2018.
- [3] G. Jocher et al., "YOLOv5," Ultralytics, 2020. [Online]. Available: <https://github.com/ultralytics/yolov5>.
- [4] G. Jocher et al., "YOLOv8: Next-generation real-time object detection," Ultralytics, 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>
- [5] R. P. Poudel, U. Bonde, S. Liwicki, and M. Zacharias, "Fast-SCNN: Fast semantic segmentation network," in Proc. BMVC, 2019.
- [6] L. C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," in Proc. IEEE CVPR, 2017.
- [7] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in Proc. MICCAI, 2015.
- [8] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in Proc. IEEE CVPR, 2018.
- [9] A. Howard et al., "Searching for MobileNetV3," in Proc. IEEE ICCV, 2019.
- [10] A. Bochkovskiy, C. Y. Wang, and H. Y. M. Liao, "YOLOv4: Optimal speed and accuracy of object detection," arXiv:2004.10934, 2020.
- [11] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in Proc. ICLR, 2021.
- [12] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," Nature, vol. 521, pp. 436–444, 2015.
- [13] M. Abadi et al., "TensorFlow: Large-scale machine learning on heterogeneous systems," 2016. [Online]. Available: <https://www.tensorflow.org>
- [14] Google LLC, "TensorFlow Lite: Machine learning for mobile and edge devices," 2020. [Online]. Available: <https://www.tensorflow.org/lite>
- [15] Google LLC, "ML Kit for mobile developers," 2023. [Online]. Available: <https://developers.google.com/ml-kit>
- [16] B. Lakde, P. Deshmukh, and S. Chavan, "Real-time obstacle detection system for visually impaired people using deep learning," IEEE Access, vol. 9, pp. 134352–134364, 2021.
- [17] A. Al-Khalifa and H. Al-Razgan, "Assistive navigation system for visually impaired using smartphone camera," Sensors, vol. 21, no. 12, pp. 1–20, 2021.
- [18] M. D. Rodríguez-Sánchez, J. Martínez-Romo, and L. M. Pardo, "Vision-based assistive navigation system for visually impaired people," Expert Systems with Applications, vol. 40, no. 18, pp. 7256–7267, 2013.
- [19] S. Gupta and R. Jain, "Deep learning based object detection system for visually impaired assistance," in Proc. IEEE Int. Conf. Artificial Intelligence and Signal Processing, 2022.
- [20] A. R. Zamir et al., "Taskonomy: Disentangling task transfer learning," in Proc. IEEE CVPR, 2018.
- [21] Flutter Team, "Flutter: A UI toolkit for building natively compiled applications," Google, 2023. [Online]. Available: <https://flutter.dev>
- [22] D. Patel and A. Shah, "Smartphone-based assistive navigation system for visually impaired users using deep learning," IEEE Access, vol. 10, pp. 86512–86525, 2022.
- [23] S. Li, B. Zhao, and Z. Zhou, "Edge AI for mobile vision applications: A survey," IEEE Internet of Things Journal, vol. 9, no. 17, pp. 15512–15530, 2022.
- [24] Y. Liu et al., "Real-time semantic segmentation for mobile devices," IEEE Transactions on Multimedia, vol. 23, pp. 4181–4193, 2021.
- [25] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in Proc. IEEE CVPR, 2017.